Database Application in Business

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OUTLINE

- Information Systems
- Data Model and Data Storage
- Application of Database

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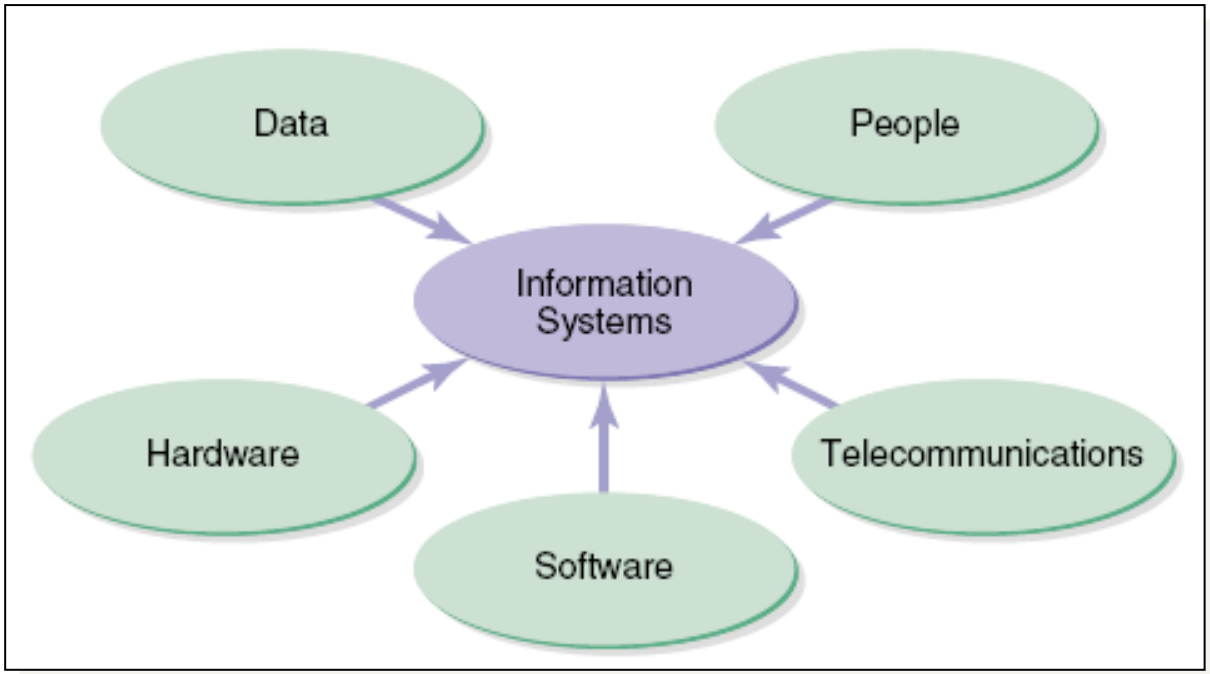
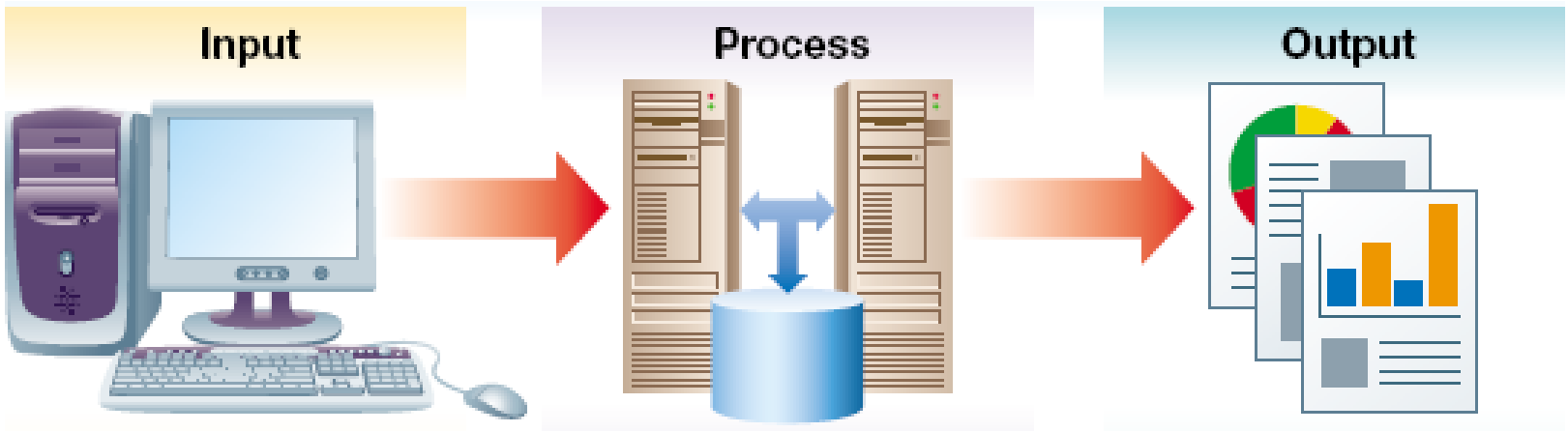
Information System

- A combination of technical components.
- Built and used by people to collect, create, and distribute useful data.
- Typically used in organizational settings but are evolving for personal use.



INFORMATION SYSTEMS

Turn Data into Information



Data

- Raw material
- Unformatted information
- Generally has no context

Information

- Processed material
- Formatted information
- Data given context

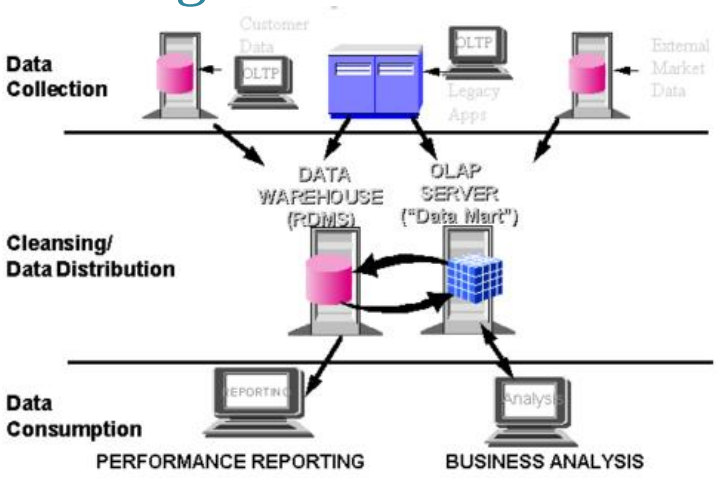
Individual time cards for factory workers entered into the payroll system	**Examples**	Department Labor Report, Project Status Report, Employee Payroll Checks

Types of Information Systems (1/2)

- **O**n**L**ine **T**ransaction **P**rocessing (OLTP)
  - an operational system for specific objective
  - massive short transactions processing
  - focus on system performance
  - emphasize the accuracy, completeness and consistency of data
  - e.g. ERP, POS, Billing, THSR online booking System ...

Types of Information Systems (2/2)

- **O**n**L**ine **A**nalytical **P**rocessing (OLAP)
  - a category of software that allows users to analyze information from **multiple database systems** at the same time.
  - performs multidimensional analysis of business data and provides the capability for *complex calculations*, *trend analysis*, and sophisticated data modeling.
  - features
    - fast (快速性)
    - analysis (分析性)
    - shared (共享性)
    - multidimensional (多維度)
    - information (資訊化)



  - e.g. data warehouse, data lake, data pool, etc.

OLTP vs. OLAP

OLTP	OLAP
• Transaction Oriented • Detailed & current data • Update/Select • Real Time update • Detail row retrieval • No redundancy • They are mostly used for – Fast online transactions – Maintain data integrity in multiple environments – Effective number of transactions per second etc.	• Subjected Oriented • Summarized and refined • Select (Read only) • Batch update • Aggregation and group by • Redundancy present • They are mostly used for – Less volume of transactions – Complex queries involving aggregations – Response time is the effective measure.

The Differences Between OLTP & OLAP

Feature	OLTP	OLAP
Function	Day to day operation	Analysis/Decision-support
Data	Current, up-to-date detailed, flat relational isolated	Historical, summarized, multidimensional integrated, consolidated
Updating/Access	Immediate/Real time Write/Read	Batch load Read-only
DB Design	Transaction-oriented (Normalization)	Subject-oriented (De-normalization)
Data Schema	Entity/Relationship	Star/Snowflake
Data Storage	Mega or Gigabytes	Tera, Peta or Zeta bytes

OUTLINE

- Types of Information Systems
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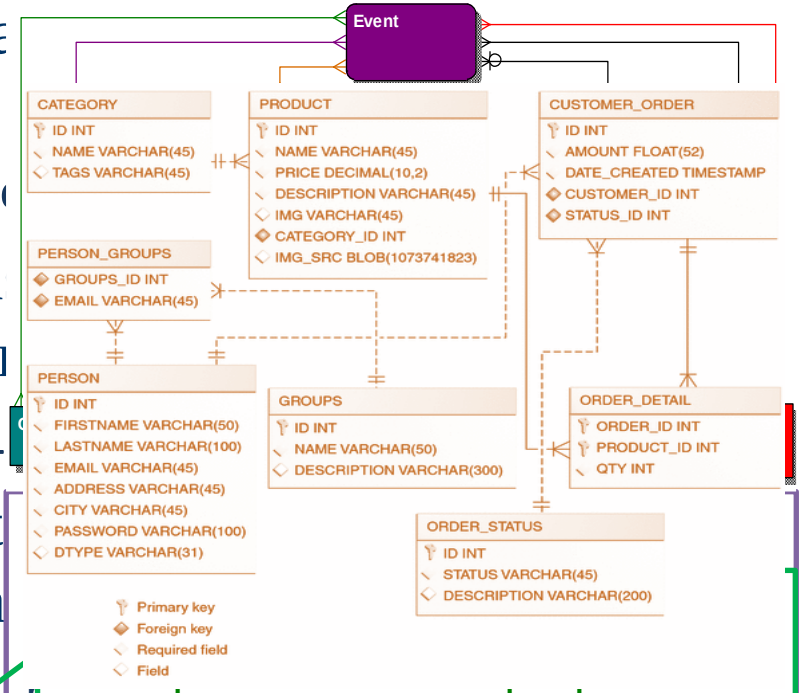
Data Model (1/2)

- Data Model
  - A set of concepts to describe the structure of a database, the operations for manipulating these structures, and certain constraints that the database should obey.
  - An abstraction of complex real-world object or event or data structures
    - Useful in understanding complexities of the real-world environment
    - Often graphical
    - Data modeling is iterative and progressive.
- Schema
  - a description of particular of collection of data, using a given data model.

Data Model (2/2)

- Components of Data Model
  - Construct is used to define the da
    elements and relationships.
    - Data Definition Language (DDL) to
  - Operations for specifying databa
    referring to concepts of the data
    - Data Manipulation Language (DML
  - Constraints specify some restrict
    constraints must be enforced at a
- Types
  - 概念資料模型 (Conceptual Data
  - 邏輯資料模型 (Logical Data Mo
  - 實體資料模型 (Physical Data Model)

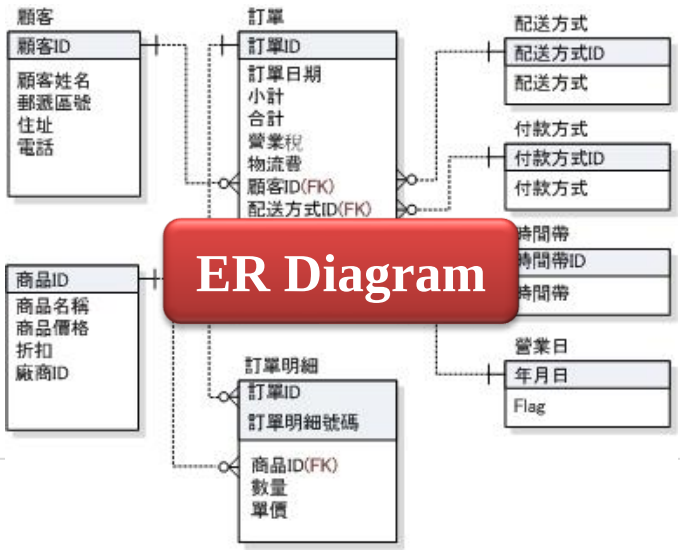


Describes **HOW** the system will be implemented. The purpose is actual implementation of the database.

Normalization vs. Denormalization

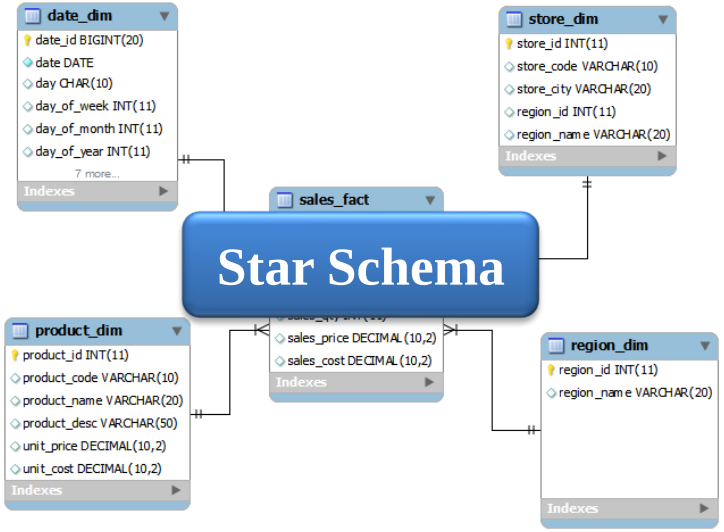
正規化 (normalization)

- The process of dividing larger tables in to smaller ones reducing the redundant data.
- To prevent databases anomalies.
- Normalize until it hurts



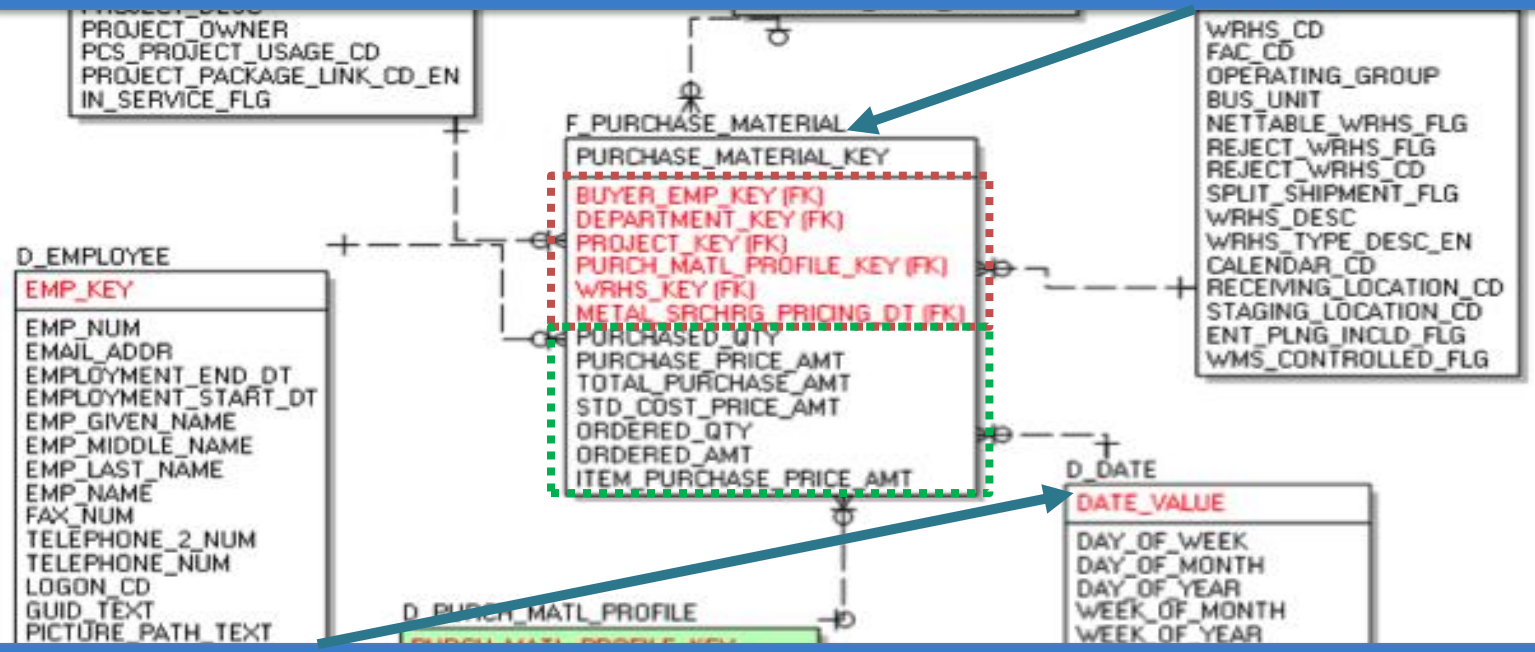
去正規化 (denormalization)

- The process of adding redundant data to optimize performance.
- To improve the read performance of the database, writes can become slower.
- Denormalize until it works.



Example of Star Schema

- A fact table is composed of two or more primary keys and usually also contains numeric data.
- It always contains at least two primary keys
- It is always a N-N relationship.

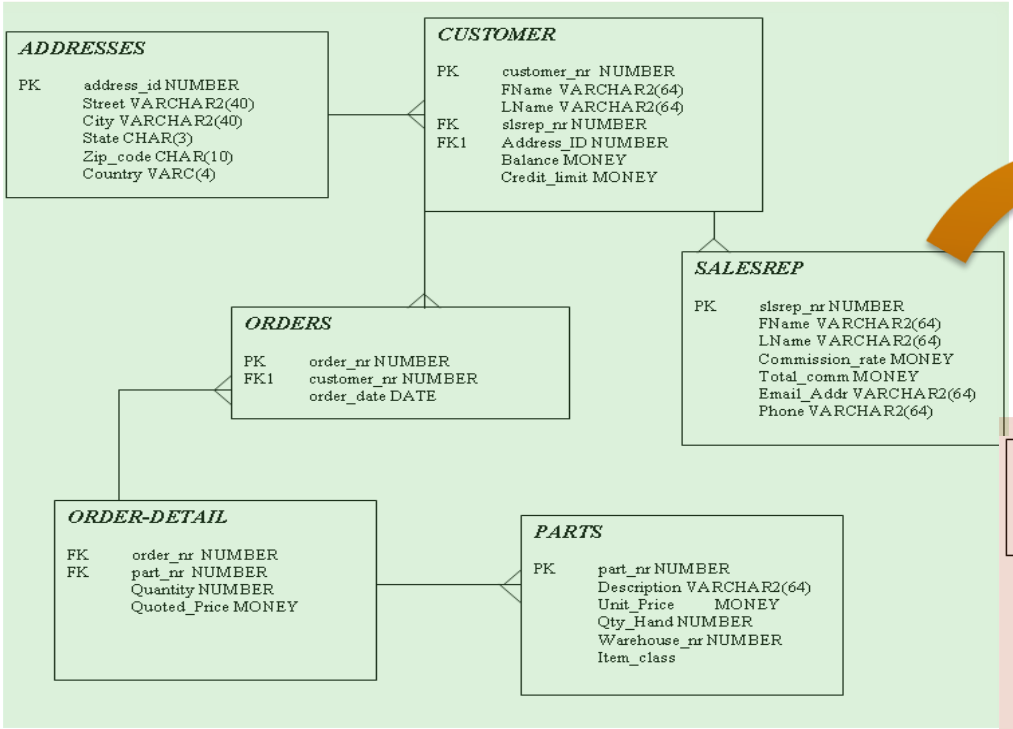


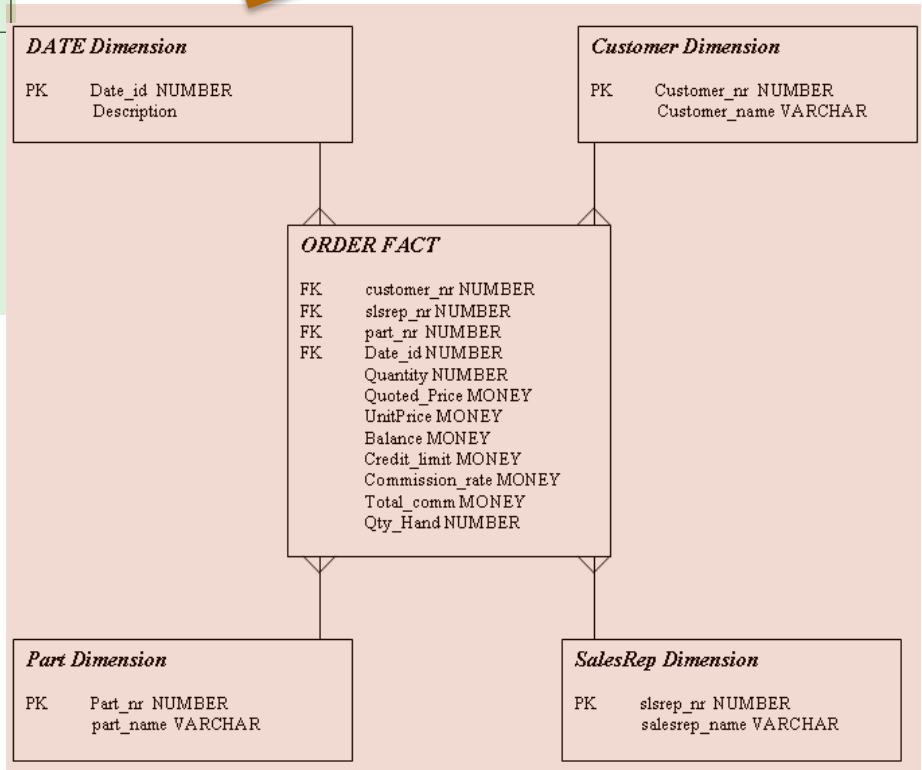
- Dimension tables is used for descriptive purposes only.
- Each dimension is considered an equal entry point into the fact table.
- The textual attributes that describe things are organized within the dimensions.

Normalization vs. De-normalization

	Normalize	De-normalization
Pro	• save most storage of all modeling techniques • simple to understand • fits well to OLTP models	• Simple design • fast queries • fits very well to OLAP models • many DBMS are optimized for queries on star schemas
Con	• bad performance with multiple joins • difficult to load (ETL) • changes are very difficult to handle	• usually centralized on one fact table • higher storage usage due to denormalization
OLTP/ OLAP	OLTP (ERP、POS, Online Shopping, …..)	OLAP (Data warehouse, Data mart, ODS…..)

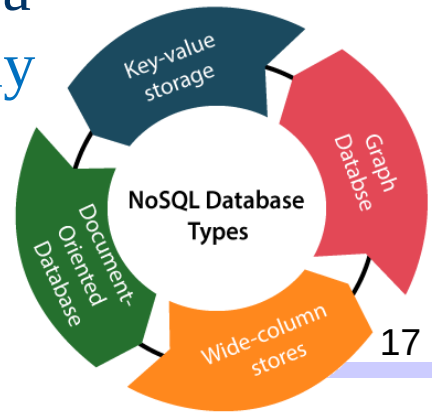
From ER-Model to Star Schema





Types of Databases (1/6)

- Relational Database (SQL)
  - This database is based on the relational data model, which stores data in the form of rows and columns, and together forms a table (relation).
  - A relational database uses SQL for storing, manipulating, as well as maintaining the data.
- NoSQL Database
  - Non-SQL/Not Only SQL is non-relational DB.
  - It is a type of database that is used for storing a wide range of data sets and stores data not only in tabular form but in several different ways.
    - Key-value(e.g. HBase), Document-oriented(e.g. MongoDB), Graph, Wide-column (e.g. Google: BigQuery, AWS: RedShift)



Types of Databases (2/6)

Relational DB

ISBN	title	author	format	price
9780992461225	JavaScript: Novice to Ninja	Darren Jones	ebook	29.00
9780994182654	Jump Start Git	Shaumik Daityari	ebook	29.00

NoSQL DB

```
{ ISBN: 9780992461225,
  title: "JavaScript: Novice to Ninja",
  author: "Darren Jones",
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}
```
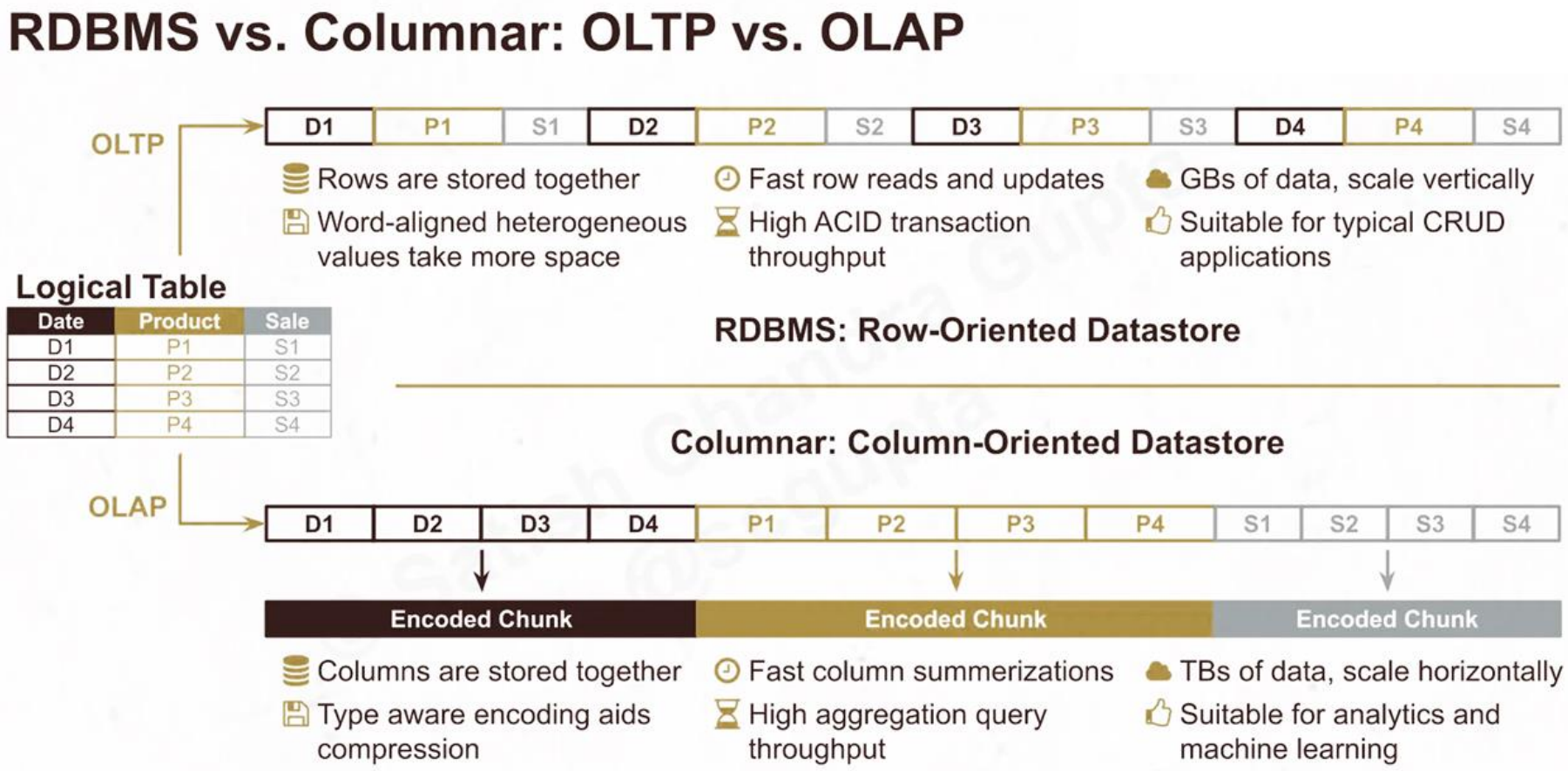

Document-Oriented NoSQL database

Types of Databases (3/6)



Types of Databases (4/6)

MongoDB vs. MySQL

NoSQL : MongoDB

document

MySQL	database	table	row	column
MongoDB	database	collection	document	field

column/field	value
first_name	robin
last_name	Jackman
title	Software Engineer
salary	3000
Internship	true

Collection (employee)

```
{
    "_id" : ObjectId("5f572f6f771ff988c1794832"),
    "first_name" : "Robin",
    "last_name" : "Jackman",
    "title" : "Software Engineer",
    "salary" : 3000,
    "Internship" : true
},
{
    "_id" : ObjectId("5f573183771ff988c1794833"),
    "first_name" : "Taylor",
    "last_name" : "Edward",
    "title" : "Software Architect",
    "salary" : 7500,
    "Internship" : false
}
```

use <dbname>

db.createCollection("collection name")

db.employee.insertOne(document, options)

Types of Databases (5/6)

MongoDB vs. MySQL

Basis	MongoDB	MySQL
Data Structure	Data is stored and represented in JSON(Javascript Object Notation) like documents.	Every record is stored as a table-cell structure with rows and column
Schema	MongoDB stores data in collections with no enforced schema. In other words, incoming data can have a pre-defined structure and it can adhere to it, however, different documents in the same collection can have different structures if required.	MySQL requires a schema definition for the tables in the databases. Therefore, the schema cannot be altered. Only inputs that conform to the given schema are accepted.
Scalability	MongoDB databases can be scaled both vertically and horizontally (horizontal scaling also known as Scaling out means adding additional nodes or machines to your infrastructure to match new demands whereas vertical scaling is adding more power to your existing machine for example upgrading CPU and RAM).	MySQL Database can be scaled vertically (Scaling vertically means adding more power to your existing machine for example upgrading CPU and RAM).
Performance	MongoDB is optimized for write performance(means writing speed performance of MongoDB database is far greater than MySQL database).	MySQL is optimized for high-performance joins across multiple tables
Handling Unstructured Data	MongoDB is an ideal choice if you have unstructured and/or structured data with the potential for rapid growth.	MySQL can not be used to handle unstructured data.

Types of Databases (6/6)

Relational DB vs. NoSQL DB

	Relational Databases	NoSQL Databases
Optimal workloads	Designed for transactional and strongly consistent online transaction processing (OLTP) applications and are good for online analytical processing (OLAP).	Designed for a number of data access patterns that include low-latency applications. NoSQL search databases are designed for analytics over semi-structured data.
Data model	The relational model normalizes data into tables that are composed of rows and columns. A schema strictly defines the tables, rows, columns, indexes, relationships between tables, and other database elements. The database enforces the referential integrity in relationships between tables.	NoSQL databases provide a variety of data models such as key-value, document, and graph, which are optimized for performance and scale.
ACID properties	Relational databases provide atomicity, consistency, isolation, and durability (ACID) properties: • Atomicity requires a transaction to execute completely or not at all. • Consistency requires that when a transaction has been committed, the data must conform to the database schema. • Isolation requires that concurrent transactions execute separately from each other. • Durability requires the ability to recover from an unexpected system failure or power outage to the last known state.	NoSQL databases often make tradeoffs by relaxing some of the ACID properties of relational databases for a more flexible data model that can scale horizontally. This makes NoSQL databases an excellent choice for high throughput, low-latency use cases that need to scale horizontally beyond the limitations of a single instance.
Performance	Performance is generally dependent on the disk subsystem. The optimization of queries, indexes, and table structure is often required to achieve peak performance.	Performance is generally a function of the underlying hardware cluster size, network latency, and the calling application.
Scale	Relational databases typically scale up by increasing the compute capabilities of the hardware or scale-out by adding replicas for read-only workloads.	NoSQL databases typically are partitionable because access patterns are able to scale out by using distributed architecture to increase throughput that provides consistent performance at near boundless scale.

Source: aws.amozon.com

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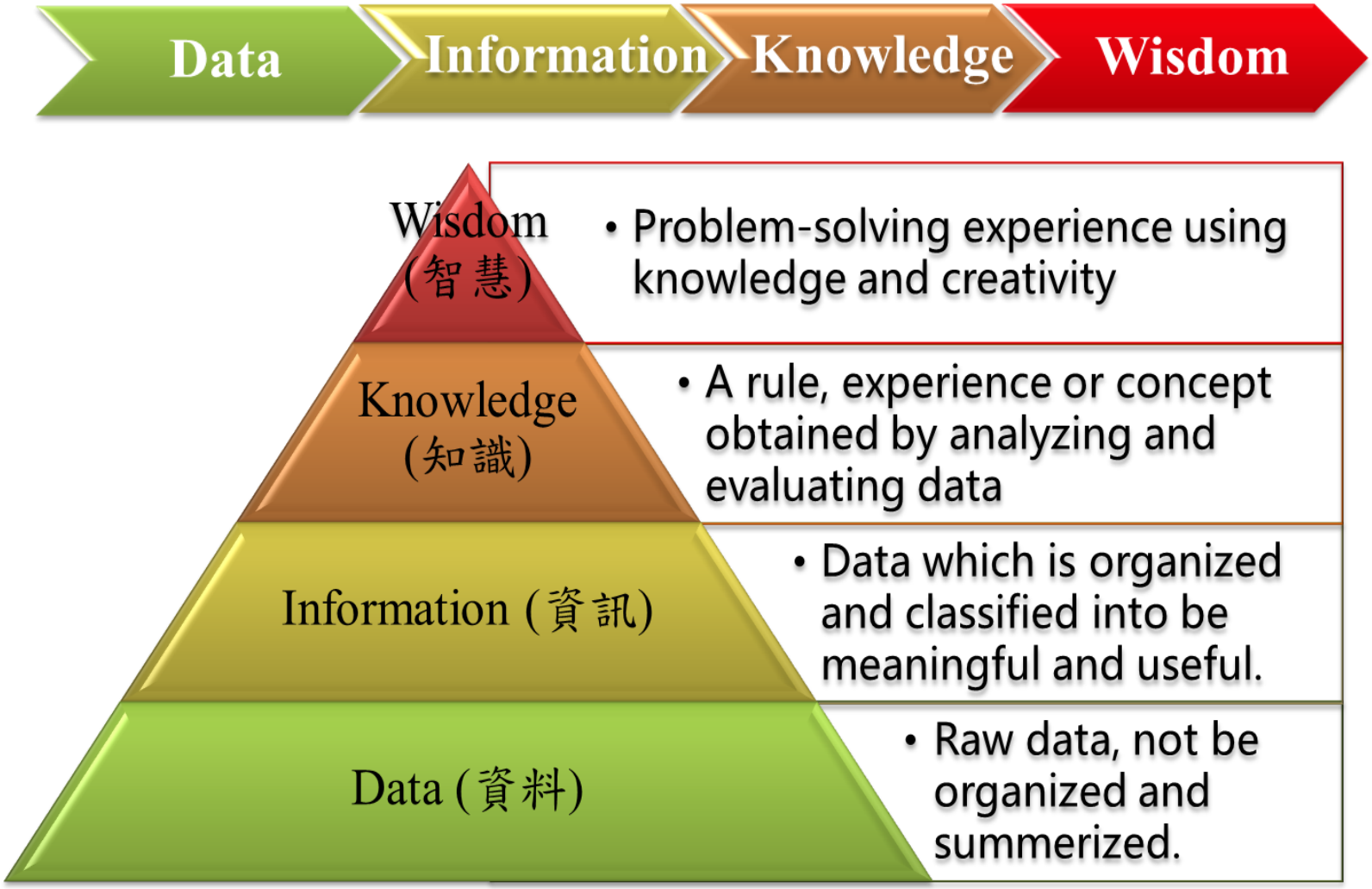
Information System Pyramid



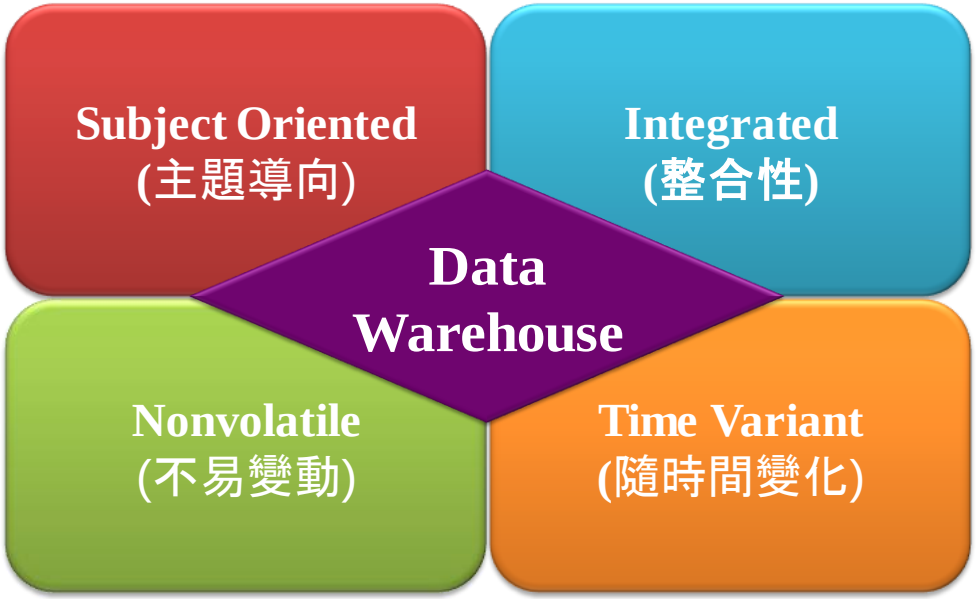
What is Data Warehouse?

- Data Warehouse is
  - simply a single, complete, and consistent store of data obtained from a variety of sources and made available to end users in a way they can understand and use it in a business context (IBM).
  - A subject-oriented, integrated, time-variant, non-updatable collection of data used in support of management decision-making processes (Bill Inmon, 1992).
  - The data repository of an enterprise. It is generally used for research and decision support.

Features of Data Warehouse

A subject oriented as it delivers information about a theme instead of organization's current operations

Integration means founding a shared entity to scale the all similar data from the different databases。

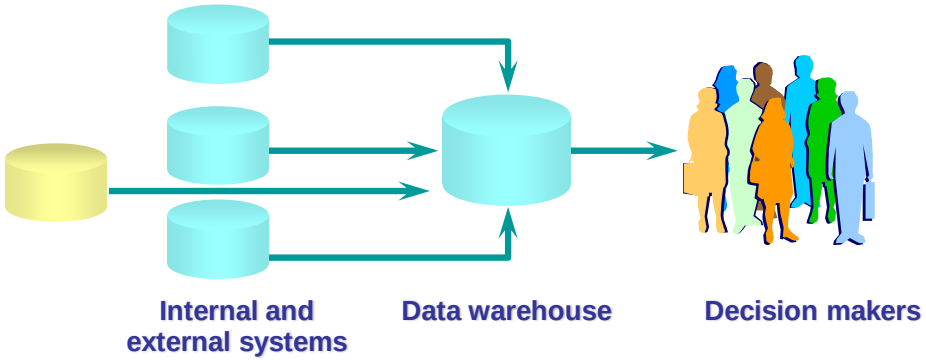


Data resided in DW is permanent. Data is not erased or deleted when new data is inserted

Data resided DW is predictable with a specific interval of time & delivers information from the historical perspective.

Functions of Data Warehouse

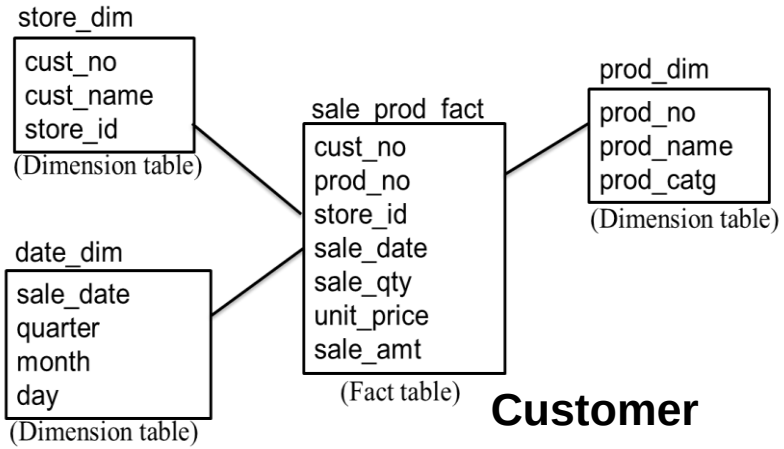
- Why need Data Warehouse?
  - Integration of information resources
  - Improve query performance
  - Support business research and decision-making
  - As a data foundation for data mining, data visualization, high-level reports and OLAP tools
- Data warehouse is used for
  - Knowledge exploration
  - Integration reports creation
  - Business insight discovery

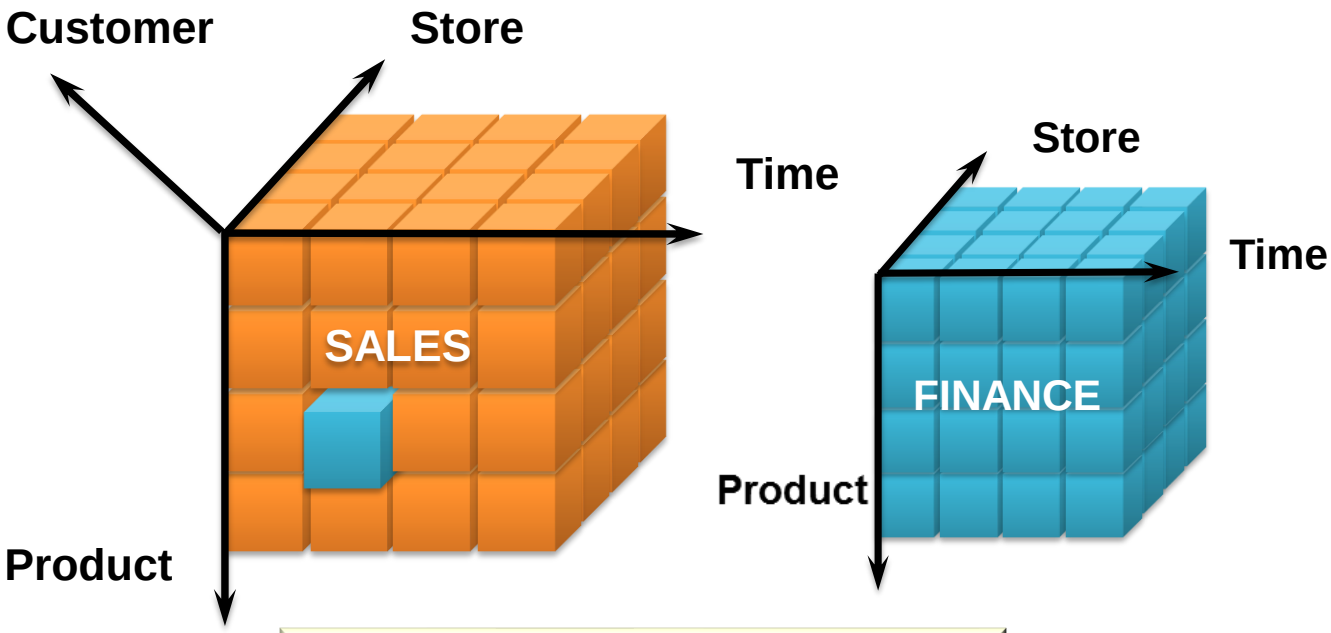


Data Structure of Data Warehouse

- Multi-dimensional database structure
- The physical data structure could be a relational database, a multi-dimensional data cube, or a mixture of the two.
- Data cube is a the multi-dimensional data structure.
  - Composed of dimensions and facts
  - Provide multi-dimensional observation of data, and aggregate data in advance for fast access to summarized data.
  - The data model should be consistent with business objective.

Multi-dimensional Data Model

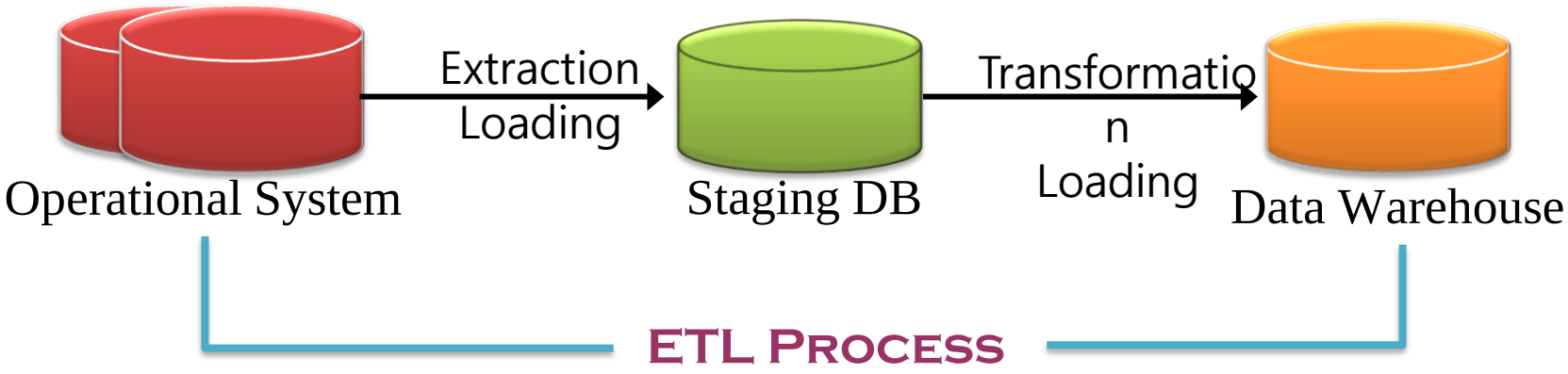




The data is found at the intersection of dimensions.

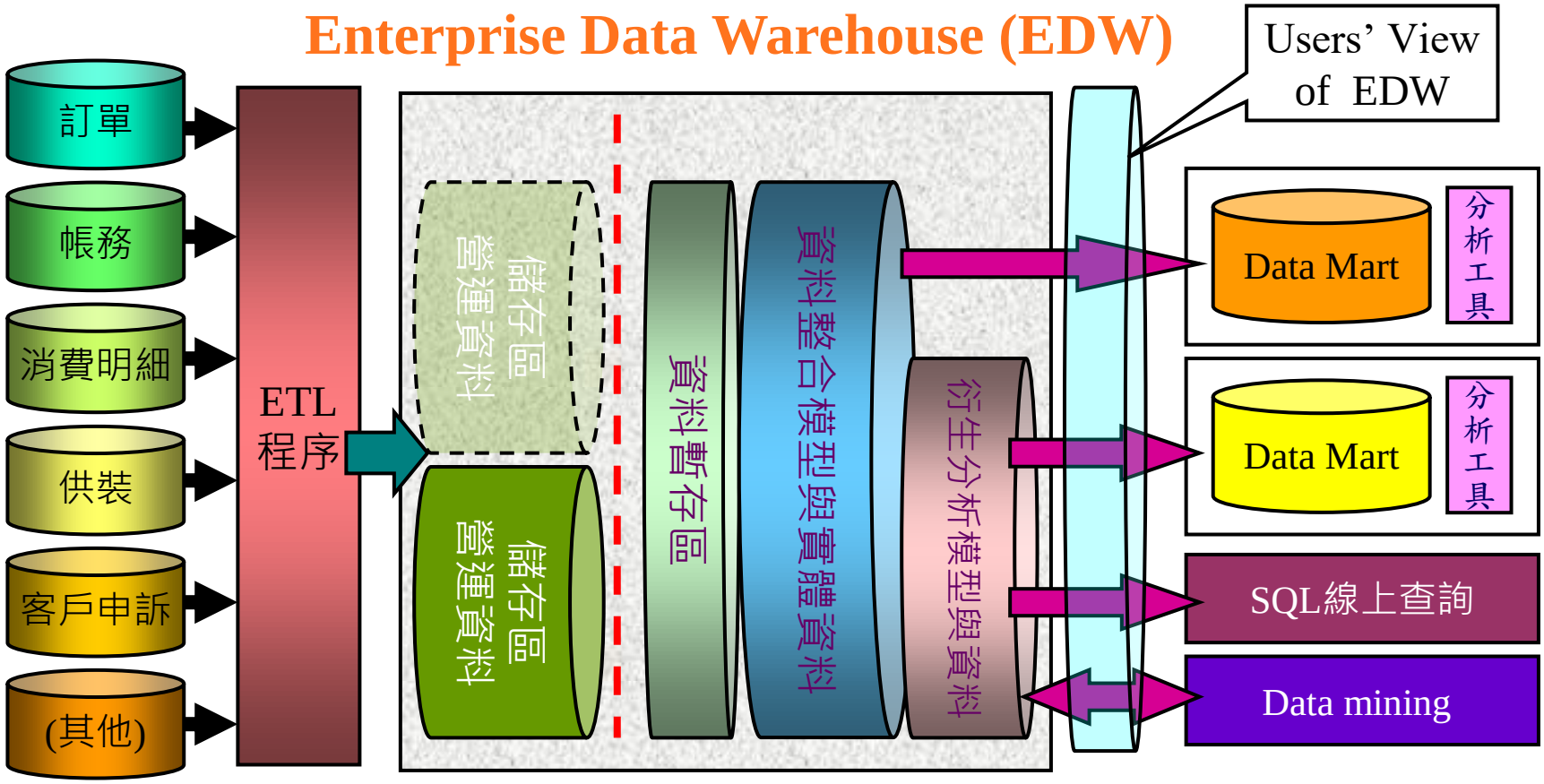
Data Process in DW

- Extraction
  - select data using different methods
- Transformation
  - validate, clean, integrate, and time stamp data
- Loading
  - move data into the warehouse



Case Study of Data Warehouse

Data Warehouse Architecture in Telecom



Data Warehouse vs. Operational DB

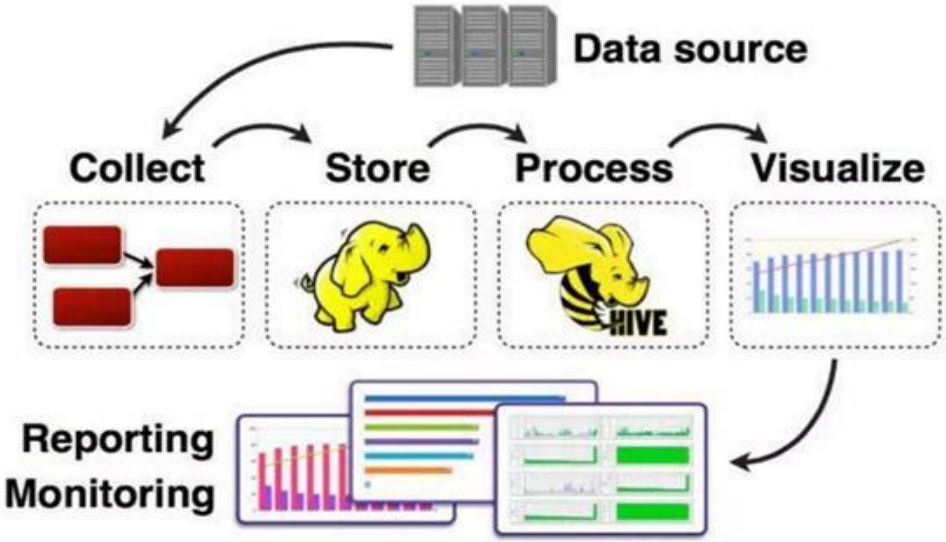
	Data Warehouse	Operational DB
Purpose	Analysis	Record
Orientation	Subject oriented	Transaction oriented
Data Type	Historic data	Current data
Data Summary	Summarized, reconciled data	Raw data
Data Model	De-normalized	Normalized
Size	Large	Small
Application	OLAP	OLTP

Large Data vs. Big Data

- There is a lot of data, you can only call it large data, not big data.
- Big Data is very big in volume, high at velocity and various types.

Big Data

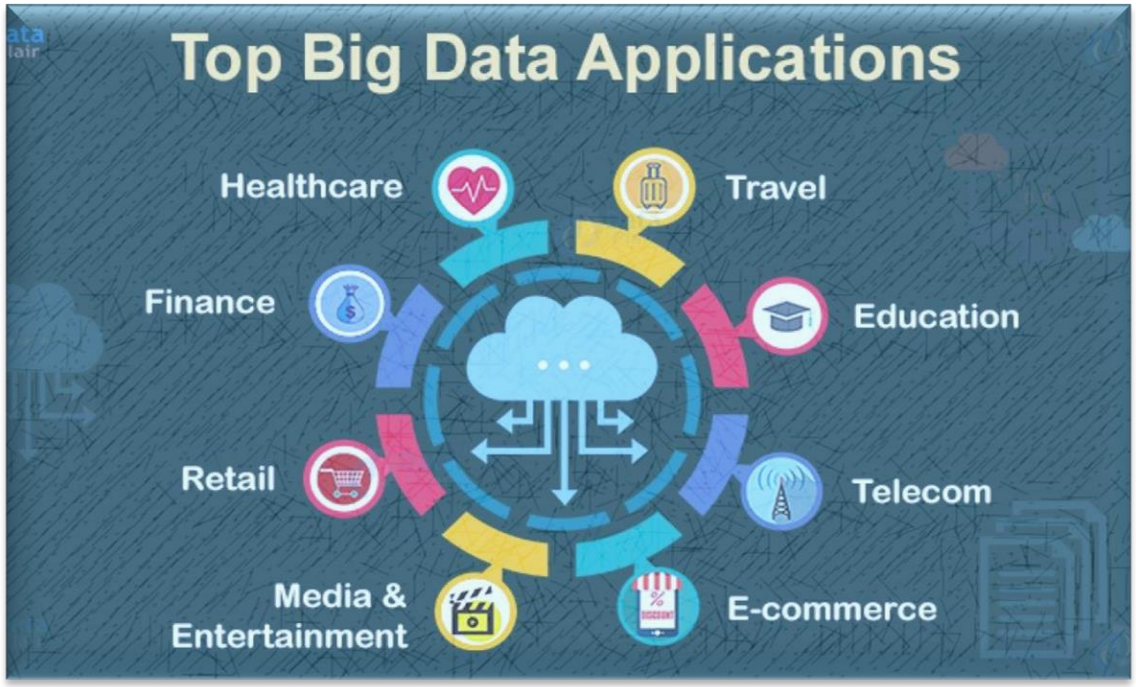
- The concept of big data originally originated in the US and was developed by Cisco, Oracle, and IBM.
- Big data is larger, more complex data sets, especially from new data sources, and can't be managed by traditional data processing software.
- These massive volumes of data can be used to address business problems you wouldn't have been able to tackle before.



大數據處理流程: 數據採集、數據存儲、數據處理、數據展現

Changes caused by Big data

Quantitative changes cause qualitative changes.
More large and more diverse but more useful!



Next Topic

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